

DATA COMMUNICATION AND COMPUTER NETWORKS

Course Code : CSE 601**Credit Units: 03****Total Hours: 45****Course Objectives:**

- The objective of the course is to provide knowledge of various multiplexing techniques and communication medias for transmission of data.
- Learn how computer network hardware and software operate.
- Investigate the fundamental issues driving network design Learn about dominant network technologies ,routing techniques ,error detection and correction techniques, various protocols and their working.

Course Contents:**Module I: Introduction: (7 Hours)**

Introduction to Data Communication, Networks-protocols, advantages, disadvantages & applications, Line Configuration, topology, Transmission mode, Classification of networks. . OSI & TCP/IP reference models, with functionality and design issues of all layers presented in the models, LAN, MAN, WAN.

Parallel & Serial Transmissions, Analog & Digital Signals, Periodic & Aperiodic Signals, Modulation-Amplitude Modulation, Frequency Modulation, Phase Modulation, Pulse Amplitude Modulation, Pulse Code Modulation.

Module II: Physical Layer: (7 Hours)

TDM, FDM, WDM; Circuit switching time division & space division switch. Transmission Media-Twisted Pair Cable, Coaxial Cable, Fiber-Optics Cable, Radio frequency Allocation, Terrestrial Microwave, Infrared rays, Satellite Communication, Cellular Telephony.

Module III: Data Link Layer: (9 Hours)

Framing, Line Discipline, Types of Errors, Error Detection & Correction (VRC, LRC, CRC, Checksum, Hamming Code), Flow Control, Error Control, CSMA/CD, Project 802, IEEE Standards-802.3, Token Bus (802.4), Token Ring (802.5).

Module IV: Network Layer: (9 Hours)

Internetworking & devices: Repeaters, Hubs, Bridges, Switches, Router, Gateway; Internet address, classful address, subnetting; Static vs. dynamic routing , Routing algorithms: shortest path algorithm, flooding, distance vector routing, link state routing; Protocols: ARP,RARP, IP, ICMP and IPV6

Module V: Transport Layer: (7 Hours)

Process to process delivery; UDP; TCP; Congestion control algorithm: Leaky bucket algorithm, Token bucket algorithm, choke packets; Quality of service: techniques to improve Qos.

Module VI: Application Layer: (6 Hours)

DNS; SMTP, SNMP, FTP, HTTP & WWW; Security: Cryptography, user authentication, security protocols in internet, Firewalls.

Advanced topics

ISDN services & ATM; DSL technology, Wireless LAN: IEEE 802.11; Bluetooth, VLAN's, Cellular telephony & Satellite network.

Course Outcomes:

- Show clear understanding of the basic concepts of data communications including the key aspects of networking and their interrelationship, packet switching, circuit switching and cell switching as internal and external operations, physical structures, types, models, and internetworking.
- Demonstrate the ability to unambiguously explain networking as it relates to the connection of computers, media, and devices (routing).
- Able to intelligently compare and contrast local area networks and wide area networks in terms of characteristics and functionalities. Able to identify limitations of typical communication systems.
- Able to differentiate among and discuss the four levels of addresses (physical, logical, port, and specific used by the Internet TCP/IP protocols).

DATA COMMUNICATION AND COMPUTER NETWORKS LAB

Course Code: CSE621**Credit Unit: 01****Total Hours: 30****Course Objective:**

To design and execute models in packet tracer to demonstrate different applications of computer networks such as addressing, routing, layer2 switching, layer3 switching, STP, VTP, VLAN, Routing tables and Routing Algorithms, RIP, EIGRP, OSPF, BGP, MPLS, EtherChannel, and Tunneling. To design and execute codes in a simulated environment to implement various Routing and Switching methods.

SOFTWARE REQUIREMENTS: Packet Tracer or GNS**Course Contents:**

Lab Experiments are based on the course Data Communication and Computer Networks (CSE601)

List of experiments/ demonstrations: (Each experiment is of 2 Hours duration)

- 11 Study of different types of cables and connectors and Network devices, display topologies demonstrating different types of devices and connectors (2 Hrs.)
- 12 Demonstrate the differences between connecting LAN and WAN devices and design LAN with WAN Topologies (2 Hrs.)
- 13 Study of basic network commands and network configuration commands (2 Hrs.)
- 14 Configure different types of VLAN using the simulator (2 Hrs.)
- 15 Configure VTP for different types of VLAN using the simulator (2 Hrs.)
- 16 Configure STP for different types of VLAN using the simulator (2 Hrs.)
- 17 Configure RSTP for different types of VLAN using the simulator (2 Hrs.)
- 18 Configure PVST, PVST+ for different types of VLAN using the simulator (2 Hrs.)
- 19 Configure RIP protocols using the simulator (2 Hrs.)
- 20 Configure RIPv6 protocols using the simulator (2 Hrs.)
- 21 Configure EIGRP protocols using the simulator (2 Hrs.)
- 22 Configure OSPF protocols using the simulator (2 Hrs.)
- 23 Configure BGP protocols using the simulator (2 Hrs.)
- 24 Configure MPLS, and GRE Tunneling using the simulator (2 Hrs.)
- 25 Configure EtherChannel, and Tunneling using the simulator (2 Hrs.)

Course Outcomes:

- Ability to identify different network topology problems and provide network design solutions.
- Ability to design and analyze different routing and switching algorithms and implement them.
- Ability to effectively use WAN and LAN design methods using simulators.

Examination Scheme:

IA			EE			
A	PR	Practical Based Test	Major Experiment	Minor Experiment	LR	Viva
5	10	15	35	15	10	10

Note: IA –Internal Assessment, EE- External Exam, A- Attendance, PR- Performance, LR – Lab Record, V – Viva.

Text & References:

- CCNA. A Comprehensive Beginners Guide To Learn About The CCNA Routing And Switching, Pearson VUE
- Cisco CCNA Command Guide, TMH

Syllabus

Programme Name: B.Tech(CSE)			Session: 2023-27
Course Code: CSE602	Course Name: Quantitative Aptitude		Semester: VI

Credits (Total)	L	T	P	Marks (Internal/External)		Contact Hours (per week)	Independent Study Hour (per week)	Section (Group)
2	2	0	0	30	70	2	2	
UG Level						Basic and applied	Student specific course outcome	Higher Education Placement Research

Course Objective:

To enhance the problem-solving skills, to improve the basic mathematical skills and to help those students who are preparing for any type of competitive examinations like CAT, CMAT, GATE, GRE, GATE, UPSC, GPSC etc. It can also help students to prepare for their placements.

Course outcomes: After completion of course, the student will be able to:

CO-1	Understand the basic concepts of QUANTITATIVE ABILITY.
CO-2	Understand the basic concepts of LOGICAL REASONING Skills
CO-3	Acquire satisfactory competency in use of VERBAL REASONING.
CO-4	Solve campus placements aptitude papers covering Quantitative Ability, Logical Reasoning and Verbal Ability.
CO-5	Compete in various competitive exams like CAT, CMAT, GATE, GRE, GATE, UPSC, GPSC etc.

Teaching Pedagogy:

T1	Classroom teaching (white board), Power Point Presentations, Interactive lectures, Inquiry based teaching
T2	ABL activities, Assignments, Flip Class/ Seminars, Quiz, Oral Viva-voce examination

Assessment Tools

AT1-1	Quiz
AT1-2	Activity Based Learning
AT1-3	Midterm Exams
AT1-4	Flip Class
AT1-5	Seminar Presentation
AT1-6	Assignments
AT1-7	Poster
AT1-8	Oral Viva-voce examination

Prerequisites: HSC Standard Mathematics, understanding of basic algebra and a familiarity with basic mathematical proficiency and ability to analyse different situations.

Module wise content details	Assessment tools
Module I: (9 Hours) Numbers System, LCM and HCF, Time & Work, Pipes and Cisterns, Time Speed Distance, Problems on Trains, Boats and Streams, Simplification and Approximation.	Quiz Mid-term Exam Assignment
Module II: (9 Hours) Percentages, Ratio Proportion and Partnership, Mixtures and Alligation, Problems on Ages, Profit and Loss, Banker's Discount, Simple Interest, Compound Interest, Clocks, Calendars.	Quiz Mid-term Exam Assignment
Module III: (9 Hours) Mensuration 2D, Mensuration 3D, Trigonometry, Height and Distances, Progressions, Logarithms, Permutation and Combination, Probability, Race,	Mid-Term Quiz Assignment
Module IV: Data Interpretation: (6 Hours) Table Charts, Bar Graph, Pie Charts, Line Charts.	Mid-Term Oral examination Seminar Presentation
Module V: Logical Reasoning: (12 Hours) Arithmetic Series: Missing Numbers, Odd One Out, Coding Decoding, Directions and Distances, Blood Relations, Data Sufficiency, Assumptions and Conclusions, Courses of Action, Puzzles, Sitting Arrangements, Syllogism, Cubes & Dice, Counting Figures.	Quiz Assignment Seminar Presentation

Additional Learning:

List of Assignments	Annexure attached
Suggested reading:	• R. S Aggarwal, "Quantitative Aptitude for competitive Exams", S. Chand publication. • Arun Sharma, "How to Prepare for Quantitative Aptitude", • R. S Aggarwal, "A Modern Approach to Verbal & Non-Verbal Reasoning", S. Chand publication. • M. Tyra, "Magical Book on Quicker Mathematics", BSC publication. •
Suggested e- resources (Websites/e-books)	• https://www.indiabix.com/aptitude/questions-and-answers/

Assessment Plan:

Component of Evaluation	Description	Code	Weightage %
Continuous Internal Evaluation	Mid Term	CT	15%
	Seminar/Viva-Voce/Quiz/Home Assignment	S/V/Q/HA	10%
Attendance	A minimum of 75% Attendance is required to be maintained by a student to be qualified for taking the End Semester examination. The dispensation of 25% includes all types of leaves including medical leaves.	A	5%
End Semester Examination	End Semester Examination	ESE	70%
Total			100%

Abbreviations: CT: Class Test, HA: Home Assignment, S/V/Q: Seminar/Viva/Quiz, ESE: End Semester Examination; A: Attendance

Course Articulation Matrix (Mapping of COs with POs)

Course Outcomes	Correlation with POs												Correlation with PSOs		
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	2	1	2	1				2		2	1			
CO2	3	3	1	3	1				2		2	1			
CO3	3	2	2	2	2				3		3	1			
CO4	3	3	2	3	2				1		2	1			
CO5	3	3	3	3	2				2		1	1			



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Syllabus

Programme Name: B. Tech. (CSE/ECE/IT)		Session: 2023-27
Course Code: CSE603	Course Name: Internet of Things	Semester: VI

Credits (Total)	L	T	P	Marks (Internal/External)		Contact Hours (per week)	Independent Study Hour (per week)	Section (Group)
3	2	1	0	30	70	3	3	
UG level						Basic and applied	Student specific course outcome	Higher Education Placement Research

Course Objective:

The objective of the course is to Introduction of IOT, Understand IOT Market perspective, Data and Knowledge Management and use of Devices in IOT Technology, Understand State of the Art – IOT Architecture. Real World IOT Design Constraints, Industrial Automation and Commercial Building Automation in IOT.

Course outcomes: After completion of course, the student will be able to:

CO-1	Understand the fundamental concept of Python in Eclipse Study, Install and Demonstrate basic operations
CO-2	Implement the different types of Arduino, Install IDE and perform basic LED programs.
CO-3	Demonstrate RFID, NFC and MQTT.
CO-4	Analyze the differences between Arduino and Raspberry Pi and Demonstrate Raspberry Pi basic LED programs and Zigbee Protocol.
CO-5	Design basic solutions using Arduino/Raspberry Pi and sensors

Teaching Pedagogy:

T1	Classroom teaching (white board), Power Point Presentations, Interactive lectures, Inquiry based teaching
T2	ABL activities, Assignments, Flip Class/ Seminars, Quiz, Oral Viva-voce examination

Assessment Tools

AT1-1	Quiz
AT1-2	Activity Based Learning
AT1-3	Midterm Exams
AT1-4	Flip Class
AT1-5	Seminar Presentation
AT1-6	Assignments
AT1-7	Poster
AT1-8	Oral Viva-voce examination
AT1-9	Industrial Visit Report

Prerequisites: HSC Standard Mathematics, knowledge of algebraic operations (manipulating equations, solving linear equations etc)

Module wise contents details	Assessment tools
Module I: Introduction to the Internet of Things: (9 Hours) Key Features, advantages, disadvantages, Wearable electronics, The Basics of Sensors & Actuators, Introduction to Cloud Computing, IOT Software.	Quiz Mid-term Exam Assignment
Module II: IoT-An Architectural Overview: (9 Hours) Building an architecture, Main design principles and needed capabilities, An IoT architecture outline, standards considerations. IoT Technology Fundamentals- Devices and gateways, Local and wide area networking, Data management, Business processes in IoT, Everything as a Service (XaaS), M2M and IoT Analytics, Knowledge Management.	Mid-Term Quiz Assignment
Module III: The Arduino Platform: (9 Hours) What is Arduino, Why Arduino, Driver installation, programming & Burning, Coding in wiring language, Compiling in Arduino, The Arduino Open-Microcontroller Platform, Arduino Basics, Arduino Board Layout & Architecture Reading from Sensors.	Mid-Term Oral Viva-voce examination Seminar Presentation
Module IV: Arduino Programming & Interface of Sensors: (9 Hours) LED display, PUSH button to array of LED, Communicating to and from computer, GSM, GPS and Zigbee interfacing, Interface sensor with arduino, Programming arduino, Reading from sensor, Connecting Arduino with Mobile Device. The Android Mobile OS, Using the Bluetooth Module.	Quiz Assignment Industrial Visit Report Seminar Presentation
Module V: Projects: (9 Hours) 1. Creating own Android App using MIT App Inventor & controlling Arduino connected devices. 2. Use Arduino to Upload free data from Environmental Sensors to Cloud Server. 3. Receive Automatic Call Notification on Mobile Phone for Burglar Alarm using IoT Platform 4. Control Electronic Devices from anywhere across the world using Internet & Mobile App.	Quiz Assignment Industrial Visit Report Poster Oral Viva-voce examination

Additional Learning:

List of Assignments	ASSIGNMENT -1 Q.1) Explain internet of things with key features, advantage and disadvantage with IOT software's. Q.2) Explain different types of IOT enabled wearable devices with specification. Q.3) Explain five smart IOT sensors and actuators with their characteristics, working principle, applications, advantage and disadvantages with their diagrams. ASSIGNMENT -2 Q.1) Describe in detail about IOT architecture with given below points: - functional block diagram - logical block diagram - IOT levels - Communication models. Q.2) Explain the given term in respect of internet of things: - data management - business processing - knowledge management ASSIGNMENT -3 Q.1) (a) Determine the importance of each layer in the generic M2M System Solution with a neat diagram. (b) Compare the pros and cons and identify different challenges of XaaS and IOT. Explain various trend in Information and communication technologies and its impact on IOT. Q.2) Illustrate the different component of Burglar Alarm. Describe the use of Burglar Alarm in various IOT applications. Calculate the cost of such system. ASSIGNMENT -4 Q.8) (a) Describe the Real-life IOT: five examples of industries and their importance of the Internet of Things for Increased Autonomy and Agility in Collaborative Production Environments? (b) Discuss the benefits and challenges of IOT during pandemic situation (COVID-19) in India. Explain how can be achieve the solution of these challenges. Q.9) Describe the various components of Arduino board and types of Arduino board with suitable diagrams. ASSIGNMENT -5 Q.1) Write short notes: - cloud computing - XaaS - Iot and M2M - Types of sensors
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	ASSIGNMENT -6 Q.1.) Explain the importance of wearable devices in healthcare system. Illustrate the types of wearable devices? Q2. Explain the different component of Burglar Alarm. Describe the use of Burglar Alarm in various IoT applications. Q3. Explain detail design guidelines of internet of thing. ASSIGNMENT -7 Q.1) Explain how to Connect Arduino with Mobile Device. The Android Mobile OS, Using the Bluetooth Module. Q.2 Creating own Android App using MIT App Inventor & controlling Arduino connected devices. ASSIGNMENT -8 Q1. Study and Configure Raspberry Pi. Q2. Explain and write the program for LED blink using Raspberry Pi. Q3. Study the Temperature sensor and Write Program for monitor temperature using Arduino ASSIGNMENT -9 Q.1) Write short notes: - Comparisons in different Arduino Board - Compare different temperature sensors Q2. Study and Implement Zigbee Protocol using Arduino / Raspberry Pi ASSIGNMENT -10 Q1. Explain the use Arduino to Upload free data from Environmental Sensors to Cloud Server. Q2. Explain working of Receive Automatic Call Notification on Mobile Phone for Burglar Alarm using IoT Platform Q3. Implement the Control Electronic Devices from anywhere across the world using Internet & Mobile App.
Suggested reading:	Text: - Jan Holler, VlasiosTsiatsis, Catherine Mulligan, Stefan Avesand, StamatisKarnouskos, David Boyle, “From Machine-to-Machine to the Internet of Things: Introduction to a New Age of Intelligence”, 1st Edition, Academic Press, 2014. Reference Books: - Vijay Madiseti and ArshdeepBahga, “Internet of Things (A Hands-on-Approach)”, 1stEdition, VPT, 2014. - Francis daCosta, “Rethinking the Internet of Things: A Scalable Approach to Connecting Everything”, 1st Edition, Apress Publications, 2013
Suggested e- resources (Websites/e-books)	https://www.arduino.cc/en/software https://www.raspberrypi.org/

Assessment Plan:

Component of Evaluation	Description	Code	Weightage %
Continuous Internal Evaluation	Mid Term	CT	15%
	Seminar/Viva-Voce/Quiz/Home Assignment	S/V/Q/HA	10%
Attendance	A minimum of 75% Attendance is required to be maintained by a student to be qualified for taking the End Semester examination. The dispensation of 25% includes all types of leaves. including medical leaves.	A	5%
End Semester Examination	End Semester Examination	ESE	70%
Total			100%

Abbreviations: CT: Class Test, HA: Home Assignment, S/V/Q: Seminar/Viva/Quiz, ESE: End Semester Examination; A: Attendance

Course Articulation Matrix (Mapping of COs with POs)

CO	CORRELATION WITH PROGRAMME OUTCOMES												CORRELATION WITH PROGRAMME SPECIFIC OUTCOMES		
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PSO3
CO1	3	2	1						1						
CO2	3	2	1						1				1		
CO3	2								1						
CO4	3	2							1						
CO5	2		1						1			1	1		

1: strongly related, 2: moderately related and 3: weakly related



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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Course Handout

Course : INTERNET OF THINGS (IOT) LAB

Course Code : CSE623, **Crédits :** 01, **Session :** 2023-27(Even Sem.), **Class :** B. Tech. 3rd Year

Introduction: The objective of the course is to Introduction of IOT, Understand IOT Market perspective, Data and Knowledge Management and use of Devices in IOT Technology, Understand State of the Art – IOT Architecture. Real World IOT Design Constraints, Industrial Automation and Commercial Building Automation in IOT.

Course Outcomes: At the end of the course, students will be able to:

CO1: Recall the fundamental concept of the Python in Eclipse Study, Install and Demonstrate basic operations.

CO2: Describe and Implement the different types of Arduino, Install IDE and perform basic LED programs.

CO3: Discuss and Demonstrate RFID, NFC and MQTT.

CO4: Compare Arduino with Raspberry Pi and Demonstrate Raspberry Pi basic LED programs and Zigbee Protocol.

Syllabus

1. Study and Install Python in Eclipse and WAP for data types in python. (3 Hours)
2. Write a Program for arithmetic operation in Python. (3 Hours)
3. Write a Program for looping statement in Python. (3 Hours)
4. Study and Install IDE of Arduino and different types of Arduino. (3 Hours)
5. Write program using Arduino IDE for Blink LED. (3 Hours)
6. Write Program for RGB LED using Arduino. (3 Hours)
7. Study the Temperature sensor and Write Program for monitor temperature using Arduino (2 Hours)
8. Study and Implement RFID, NFC using Arduino. (2 Hours)
9. Study and implement MQTT protocol using Arduino. (2Hours)
10. Study and Configure Raspberry Pi. (2 Hours)
11. WAP for LED blink using Raspberry Pi. (2 Hours)
12. Study and Implement Zigbee Protocol using Arduino / Raspberry Pi. (2Hours)

Examination Scheme:

IA			EE			
A	PR	Practical Based Test	Major Experiment	Minor Experiment	LR	Viva
5	10	15	35	15	10	10

Note: IA – Internal Assessment, EE - External Exam, A - Attendance, PR- Performance, LR – Lab Record, V – Viva.

Lab Plan

Lab Session	Program Name	Mapped CO	Mode of Assessing CO
1	Study and Install Python in Eclipse and WAP for data types in python.	CO1	Viva/Quiz & End Sem Practical Exam
2	Write a Program for arithmetic operation in Python.	CO1	Viva/Quiz & End Sem Practical Exam
3	Write a Program for looping statement in Python.	CO1	Viva/Quiz & End Sem Practical Exam
4	Study and Install IDE of Arduino and different types of Arduino.	CO2	Viva/Quiz & End Sem Practical Exam
5	Write program using Arduino IDE for Blink LED.	CO2	Viva/Quiz & End Sem Practical Exam
6	Write Program for RGB LED using Arduino.	CO2	Viva/Quiz & End Sem Practical Exam
7	Study the Temperature sensor and Write Program for monitor temperature using Arduino	CO2	Viva/Quiz & End Sem Practical Exam
8	Explain working and WAP to implement ultrasonic sensor using Arduino.	CO2	Viva/Quiz & End Sem Practical Exam
9	Study and Implement RFID/ NFC using Arduino.	CO3	Viva/Quiz & End Sem Practical Exam
10	Study and implement MQTT protocol using Arduino.	CO3	Viva/Quiz & End Sem Practical Exam
11	Study and Implement Zigbee Protocol using Arduino.	CO3	Viva/Quiz & End Sem Practical Exam
12	Study and Configure Raspberry Pi.	CO4	Viva/Quiz & End Sem Practical Exam
13	WAP for LED blink using Raspberry Pi.	CO4	Viva/Quiz & End Sem Practical Exam
14	Explain working and WAP to implement ultrasonic sensor using Raspberry Pi..	CO4	Viva/Quiz & End Sem Practical Exam
15	Study and Implement Zigbee Protocol using Raspberry Pi.	CO4	Viva/Quiz & End Sem Practical Exam

16. Course Articulation Matrix (Mapping of COs with POs)

CO	CORRELATION WITH PROGRAMME OUTCOMES												CORRELATION WITH PROGRAMME SPECIFIC OUTCOMES		
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	1	1	1	2				1		1	1			
CO2	3	2	1	2	2				1		1	1			
CO3	3	2	2	2	2				1		1	1			
CO4	3	2	2	2	2				1		1	1			

1: strongly related, 2: moderately related and 3: weakly related

PROBLEM SOLVING TECHNIQUES

Course Code: CSE 604**Credit Units: 03
Total Hours: 45****Course Objective:**

To improve problem solving skills using the concept of C,C++ and data structures and develop knowledge of basic data structures for storage and retrieval of ordered or unordered data. Data structures include: arrays, linked lists, binary trees, heaps, and hash tables etc.

Course Contents:**Module I: Programming in C –I: (7 Hours)**

Introduction: Basic structure of C program, Concept of variables, constants and data types in C, Operators and expressions: Introduction, arithmetic, relational, Logical, Assignment, Increment and decrement operator, Conditional, bitwise operators, Expressions, Operator precedence and associativity. Managing Input and output Operation, formatting I/O.

Control Structures and Looping: C Statements, conditional executing using if, else, nesting of if, switch and break Concepts of loops, example of loops in C using for, while and do-while, continue and break. Storage types (automatic, register etc.), predefined processor, Command Line Argument.

Module II: Programming in C – II: (9 Hours)

Arrays and Functions: One dimensional arrays and example of iterative programs using arrays, 2-D arrays Use in matrix computations. Concept of Sub-programming, functions Example of user defined functions. Function prototype, Return values and their types, calling function, function argument, function with variable number of argument, recursion.

Pointers: Pointers, relationship between arrays and pointers Argument passing using pointers, Array of pointers. Passing arrays as arguments.

String: Strings and C string library.

Structure: Structure and Union, Defining C structures, Giving values to members, Array of structure, Nested structure, passing strings as arguments.

File Handling: Basics of file Handling.

Module III: Object Oriented Programming in C++: (8 Hours)

Difference between C and C++, Procedure Oriented and Object Oriented Approach, Characteristics of Object-Oriented Languages

Classes and Objects: Abstract data types, Object & classes, attributes, methods, C++ class declaration, Local Class and Global Class, State identity and behaviour of an object, Local Object and Global Object, Scope resolution operator, Friend Functions, Inline functions, Constructors and destructors, instantiation of objects, Types of Constructors, Static Class Data, Array of Objects, Constant member functions and Objects, Memory management Operators.

Inheritance: Inheritance, Types of Inheritance, access modes – public, private & protected, Abstract Classes, Ambiguity resolution using scope resolution operator and Virtual base class, Aggregation, composition vs classification hierarchies, Overriding inheritance methods, Constructors in derived classes, Nesting of Classes.

Polymorphism: Polymorphism, Type of Polymorphism – Compile time and runtime, Function Overloading, Operator Overloading (Unary and Binary) Polymorphism by parameter, Pointer to objects, this pointer, Virtual Functions, pure virtual functions.

Module IV: Data Structure –I: (10 Hours)

Classification of Data structures, Abstract Data Types, Implementation aspects: Memory representation. Data structures operations and its cost estimation.

Linked List: Representation of linked list in memory, different implementation of linked list. Circular linked list, doubly linked list, etc. Application of linked list: polynomial manipulation using linked list, etc.

Stacks: Stacks as ADT, Different implementation of stack, multiple stacks. Application of Stack: Conversion of infix to postfix notation using stack, evaluation of postfix expression, Recursion.

Queues: Queues as ADT, Different implementation of queue, Circular queue, Concept of Dqueue and Priority Queue, Application of queues.

Module V: Data Structure-II: (11 Hours)

Tree: Definitions - Height, depth, order, degree etc. Binary Search Tree - Operations, Traversal, Search. AVL Tree, Heap, Applications and comparison of various types of tree; Introduction to forest, multi-way Tree, B tree, B+ tree, B* tree and red-black tree.

Graphs: Introduction, Classification of graph: Directed and Undirected graphs, etc, Representation, Graph Traversal: Depth First Search (DFS), Breadth First Search (BFS), Graph algorithm: Minimum Spanning Tree (MST)- Kruskal, Prim's algorithms. Dijkstra's shortest path algorithm; Comparison between different graph algorithms. Application of graphs.

Sorting: Introduction, Sort methods like: Bubble Sort, Quick sort. Selection sort, Heap sort, Insertion sort, Shell sort, Merge sort and Radix sort; comparison of various sorting techniques. Basic Search Techniques: Sequential search, Binary search, Comparison of search methods, Hashing & Indexing.

Course outcomes:

- Able to understand the concepts of data structure, data type and array data structure.
- Able to implement linked list data structure to solve various problems.
- Able to understand and apply various data structure such as stacks, queues, trees and graphs to solve various computing problems using C/C++ -programming language.
- To apply concepts and techniques for implementation.

Examination Scheme:

Components	A	CT	S/V/Q/HA	ESE
Weightage (%)	5	15	10	70

A: Attendance, CT: Class Test, S/V/Q/HA: Seminar/Viva/Quiz/ Home Assignment, ESE: End Semester Examination;

Text & References:

Text:

- Yashwant Kanetkar, "Let us C", BPB Publications, 2nd Edition, 2001.
- Herbert Schildt, "C: The complete reference", Osbourne Mcgraw Hill, 4th Edition, 2002.
- A.R. Venugopal, Rajkumar, T. Ravishanker "Mastering C++", TMH, 1997
- R. Lafore, "Object Oriented Programming using C++", BPB Publications, 2004.
- A. M. Tenenbaum, "Data Structures using C & C++", Prentice-Hall of India Pvt. Ltd., New Delhi.

Reference:

- Kernighan & Ritchie, "C Programming Language", The (Ansi C Version), PHI, 2nd Edition.
- "Object Oriented Programming with C++" By E. Balagurusamy.
- Bruno R Preiss, "Data Structures and Algorithms with Object Oriented Design Pattern in C++", Jhon Wiley & Sons, Inc.
- Gilberg Forozan, "Data Structure – A pseudo code approach with C++", Cengage Learning, New Delhi.

PROBLEM SOLVING TECHNIQUES LAB**Course Code: CSE 624****Credit Units: 02****Total Hours: 60****Course Objective:**

To write the programs for solving problems using the concept of C,C++ and data structures and develop knowledge of basic data structures for storage and retrieval of ordered or unordered data.

Software Requirements: Turbo C++ compiler

Course Contents :

Lab Experiments are based on the course Problem Solving Techniques (CSE 604)

List of experiments/demonstrations:**(A) Programming in C : (20 Hours)**

1. Write a simple program based on operators (pre, post increment , bitwise and , or , etc.).
2. Write a simple program based on conversions (from int to float & float to int)
3. Write a program for find the max and min from the three numbers.
4. Write the program for the simple, compound interest.
5. Write program for students marks grading.
6. C program to check whether a given number is odd or even.
7. C program to Add digits of input number.
8. C program to Factorial of a given number.
9. C program to swap two numbers without using third variable.
10. C program to check whether a given year is leap year or not.
11. C program to check whether a given number Palindrome Number or not.
12. C programs to print different patterns.
13. Program for the following using switch statement:

Menu:-

- (a) Sum of two numbers
 - (b) Negative or Positive Number
 - (c) Simple Interest
 - (d) Area of Circle
 - (e) Exit
14. C program to check whether a given number Prime Number or not.
 15. C program to check whether a given number Armstrong Number or not.
 16. C program to print Fibonacci series up to given term.
 17. C program to find out sum of 10 numbers by using array.
 18. C program to reverses of one array elements into another.
 19. C program to find out maximum and minimum number in an array.
 20. Write a C program that uses functions to perform the following:
 - (a) Addition of Two Matrices
 - (b) Multiplication of Two Matrices
 - (c) Transpose of a matrix
 21. C program to Factorial of a given number by using user define function.
 22. Write a program for display values reverse order from array using pointer.
 23. Write a program through pointer variable to sum of n elements from array .
 24. Write a C program which copies one file to another.

(B) Object Oriented Programming in C++ : (20 Hours)

1. Write a program that show the concept of class and object and having function for addition, subtraction, multiplication and division of two number.

2. Program that show the concept of inline function.
3. Program that show the concept of friend function.
4. Program that show the concept of all types of constructor and destructor.
5. Program that show the concept of local class and global class.
6. Program that show the concept of local object and global object.
7. Program that show the concept of static class data and static member function.
8. Program that show the concept of constant member data and function .
9. Program that show the concept of dynamic memory allocation.
10. Program that show the concept of multiple inheritance.
11. Program that show the concept of multilevel inheritance.
12. Program that show the concept of function overloading.
13. Program that show the concept of function overriding.
14. Program that illustrates the order of execution of constructors and destructors when new class is derived from more than one base class.
15. Program that show the concept of operator overloading(overload ++ operator) .
16. Program that overload +,- for addition and subtraction of two complex number.
17. Program that show the concept of this pointer.
18. Program that illustrates how run time polymorphism is achieved using virtual functions.
19. Program that illustrates the role of virtual base class in building class hierarchy.
20. Program that illustrates the role of abstract class in building class hierarchy.

(C) Data Structure : (30 Hours)

1. Write a C/C++ program that uses functions to perform the following: i) Create a singly linked list of integers. ii) Delete a given integer from the above linked list. iii) Display the contents of the above list after deletion.
2. Write a C/C++ program that uses functions to perform the following: i) Create a doubly linked list of integers. ii) Delete a given integer from the above doubly linked list. iii) Display the contents of the above list after deletion.
3. Write a C/C++ program that implement the concept of Stack using array/link list.
4. Write a C/C++ program that implement the concept of Queue using array/link list..
5. Write a C/C++ program that implement the concept of Circular Queue.
6. Write a C/C++ program that implement the solution of Tower of Hanoi problem.
7. Write a C/C++ program that uses stack operations to convert a given infix expression into its postfix Equivalent.
8. Write a C/C++ program that uses functions to perform the following: i) Create a binary search tree of characters. ii) Traverse the above Binary search tree recursively in postorder.
9. Write a C/C++ program that uses functions to perform the following: i) Create a binary search tree of integers. ii) Traverse the above Binary search tree non recursively in order.
10. Write C/C++ programs for implementing the following sorting methods to arrange a list of integers in ascending order: i) Insertion sort ii) Bubble Sort iii) Insertion Sort iv) Quick Sort v) Merge sort vi) Counting Sort etc.
11. Write C/C++ programs for implementing the following graph traversal algorithms:
(i)Depth first traversal (ii)Breadth first traversal

Course outcomes:

Able to write the program using different data structures.

- Able to implement linked list data structure to solve various problems.
- Able to apply various data structure such as stacks, queues, trees and graphs to solve various computing problems using C/C++ -programming language.
- To apply concepts and techniques for implementation.

Examination Scheme:

IA	EE
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SOFTWARE ENGINEERING

Course Code: CSE 605**Credit Units: 03****Total Hours: 45****Course Objective:**

The basic objective of Software Engineering is to develop methods and procedures for software development that can scale up for large systems and that can be used to consistently produce high-quality software at low cost and with a small cycle time. Software Engineering is the systematic approach to the development, operation, maintenance, and retirement of software.

The course provides a thorough introduction to the fundamentals principles of software engineering. The organization broadly is based on the classical analysis-design-implementation framework.

Course Contents:**Module I: Introduction: (9 Hours)**

Software life cycle models: Waterfall, Prototype, Evolutionary and Spiral models, Overview of Quality Standards like ISO 9001, SEI-CMM.

Module II: Software Metrics and Project Planning: (9 Hours)

Size Metrics like LOC, Token Count, Function Count, Design Metrics, Data Structure Metrics, Information Flow Metrics. Cost estimation, static, Single and multivariate models, COCOMO model, Putnam Resource Allocation Model, Risk management.

Module III: Software Requirement Analysis, design and coding: (10 Hours)

Problem Analysis, Software Requirement and Specifications, Behavioural and non-behavioural requirements, Software Prototyping Cohesion & Coupling, Classification of Cohesiveness & Coupling, Function Oriented Design, Object Oriented Design, User Interface Design Top-down and bottom-up Structured programming, Information hiding.

Module IV: Software Reliability, Testing and Maintenance: (11 Hours)

Failure and Faults, Reliability Models: Basic Model, Logarithmic Poisson Model, Software process, Functional testing: Boundary value analysis, Equivalence class testing, Decision table testing, Cause effect graphing, Structural testing: path testing, Data flow and mutation testing, unit testing, integration and system testing, Debugging, Testing Tools, & Standards. Management of maintenance, Maintenance Process, Maintenance Models, Reverse Engineering, Software RE-engineering.

Module V: UML: (6 Hours)

Introduction to UML, Use Case Diagrams, Class Diagram: State Diagram in UML Activity Diagram in UML Sequence Diagram in UML Collaboration Diagram in UML.

Course Outcomes:

- Ability to use the modeling approaches for the designing of software.
- Ability to use the testing tools and designing of test cases for testing.
- Ability to use the Unified modeling language (UML) for the designing of software product.

Examination Scheme:

Components	A	CT	S/V/Q/HA	ESE
Weightage (%)	5	15	10	70

A: Attendance, CT: Class Test, S/V/Q/HA: Seminar/Viva/Quiz/ Home Assignment, ESE: End Semester Examination;

Text & References:**Text:**

- K. K. Aggarwal & Yogesh Singh, "Software Engineering", 2nd Ed, New Age International, 2005.
- R. S. Pressman, "Software Engineering – A practitioner's approach", 5th Ed., McGraw Hill Int. Ed., 2001.

References:

- R. Fairley, "Software Engineering Concepts", Tata McGraw Hill, 1997.
- P. Jalote, "An Integrated approach to Software Engineering", Narosa, 1991.



AMITY UNIVERSITY

MADHYA PRADESH

Established vide Government of Madhya Pradesh Act No. 27 of 2010

Syllabus

Programme Name: B. Tech. (CSE)		Session: 2023-27
Course Code: IT601	Course Name: COMPUTER NETWORKS	Semester: VI

Credits (Total)	L	T	P	Marks (Internal/External)		Contact Hours (per week)	Independent Study Hour (per week)	Section (Group)
3	2	1	0	30	70	3	3	
UG level						Basic and applied	Student specific course outcome	Higher Education Placement Research

Course Objective:

The objective of the course is to provide introductory concepts in Communications such as Signaling, Encoding, Modulation, Error Detection & Correction. The course is also aimed at providing basic understanding of Computer networks starting with OSI Reference Model, Protocols at different layers with special emphasis on IP, TCP & UDP and Routing algorithms. Students are expected to learn applications of Computer Networks and interfacing standards along with current edge of the network security techniques.

Course outcomes: After completion of course, the student will be able to:

CO-1	Show clear understanding of the basic concepts of data communications including the key aspects of networking and their interrelationship, packet switching, circuit switching and cell switching as internal and external operations, physical structures, types, models, and internetworking.
CO-2	Able to intelligently compare and contrast local area networks and wide area networks in terms of characteristics and functionalities. Able to identify limitations of typical communication systems.
CO-3	Able to differentiate among and discuss the four levels of addresses (physical, logical, port, and specific used by the Internet TCP/IP protocols.
CO-4	Able to understand applications and security issues of computer networks

Teaching Pedagogy:

T1	Classroom teaching (white board), Power Point Presentations, Interactive lectures, Inquiry based teaching
T2	Class Test, Assignments, Flip Class/ Seminars, Quiz, Oral Viva-voce examination

Assessment Tools

AT1-1	Quiz/Class Test
AT1-2	Activity Based Learning
AT1-3	Midterm Exams
AT1-4	Flip Class
AT1-5	Seminar Presentation
AT1-6	Assignments
AT1-7	Poster
AT1-8	Oral Viva-voce examination
AT1-9	Industrial Visit Report

Prerequisites: computer science fundamentals, networking basics, mathematics (especially discrete mathematics), digital logic, operating systems, programming skills, and basic electronics knowledge

Module wise contents details	Assessment tools
Module I: Introduction to Data Communications (9 Hours) Introduction to Data Communication, Networks-protocols, advantages, disadvantages & applications, Line Configuration, topology, Transmission mode, Classification of networks. OSI and TCP IP Model. Parallel & Serial Transmissions, Analog & Digital Signals, Periodic & Aperiodic Signals, Modulation---Amplitude Modulation, Frequency Modulation, Phase Modulation, Pulse Amplitude Modulation, Pulse Code Modulation, Sampling.	Quiz/Class Test Mid-term Exam Assignment
Module II: Data Link Layer (9 Hours) Framing, Line Discipline, Types of Errors, Error Detection & Correction (VRC, LRC, CRC, Checksum, Hamming Code), Flow Control (Stop-and-wait & Sliding Window), Error Control (Stop & Wait ARQ, Sliding Window ARQ using Go-back n method and Selective-Reject). CSMA/CD, Project 802, IEEE Standards-802.3, Token Bus (802.4), Token Ring (802.5), Introduction to Bridges.	Quiz/Class Test Mid-term Exam Assignment
Module III: Network and Transport Layer (9 Hours) Network Layer in Internet-IP protocol, IP addressing, Subnetting & Internet Control Protocols, Transport Layer in Internet-TCP & UDP, Internal Organization of Network Layer, Routing Algorithms-Shortest Path Routing, Flooding, Distance Vector Routing, Link State Routing, Transport Layer Design issues, General Principles of Congestion, Congestion Prevention Policies, Duties of Transport Layer, Connection Establishment & Connection Termination.	Quiz/Class Test Mid-term Exam Oral Viva-voce examination Seminar
Module IV: Application Layer (9 Hours) Application Layer Protocols – Domain name space, DNS in internet, web server, electronic mail, FTP, WWW, HTTP, SMTP, POP, IMAP, SNMP, DHCP, NFS, NTP, Telnet, SSH.	Quiz/Class Test Mid-term Exam Oral Viva-voce examination Presentation
Module V: Network Security (9 Hours) Introduction to the Concept of Security, SSL, Cryptographic Techniques in Network Security, Computer-based Symmetric and Asymmetric Key Cryptographic Algorithms, Public Key Infrastructure (PKI), Internet Security Protocols, Network Security design paradigm.	Quiz Assignment Industrial Visit Report Poster Oral Viva-voce examination

Additional Learning:

List of Assignments	• Explain Http and WWW in details. • Differentiate OSI and TCP/IP. • Differentiate IPV4 and IPV6. • What are the types of cryptography explain with diagram? • Explain Hamming Code in data link layer • Differentiate unicast, multicast, anycast in IPV6 • Explain the Techniques to improve QoS. • Explain the flow control techniques in data link layer with suitable diagram. • What is DNS? What is the need of DNS explain with its working diagram? • Explain the ISDN services and ATM Concept. • Differentiate LAN and VLAN • Differentiate IEEE 802.11 and IEEE 802.15a • What do you understand by Cellular telephony and Satellite network? • Explain the concept of SONET.
Suggested reading:	• Behrouz., Forouzan., “Data Communication and Networking”, TMH • W. Stallings, "Data and Computer Communication" PHI • A.S. Tanenbaum, "Computer Networks" , PHI • Kennedy, “Electronics Communication System”, TMH. • Kurose and Rose – “Computer Networking -A top down approach featuring the internet” – Pearson Education • Leon, Garica, Widjaja – “Communication Networks” – TMH • Walrand – “Communication Networks” – TMH.
Suggested e-resources (Websites/e-books)	3. Online Courses and Tutorials: • Coursera: "Computer Networking" by Stanford University. • edX: "Introduction to Computer Networking" by NYU. • Udemy: Various courses on networking fundamentals and advanced topics. • NPTEL: Various courses on networking fundamentals and advanced topics. 4. Books: • "Computer Networking: A Top-Down Approach" by James F. Kurose and Keith W. Ross. • "Data Communications and Networking" by Behrouz A. Forouzan. • "TCP/IP Illustrated" by Kevin R. Fall and W. Richard Stevens. 5. Websites and Documentation: • Cisco Networking Academy: Offers various free resources, including courses and interactive learning tools. • Wireshark: A network protocol analyzer that can be used for learning and analyzing network traffic. • Internet Engineering Task Force (IETF): Provides RFCs (Request for Comments) which are documents that define the standards and protocols used in computer networking. • The TCP/IP Guide: An online resource providing comprehensive information on TCP/IP protocols and networking concepts.

	6. Online Tools: • Packet Tracer: A network simulation tool developed by Cisco for designing, configuring, and troubleshooting network scenarios. • GNS3 (Graphical Network Simulator-3): A network emulator that allows users to simulate complex networks. 7. Online Forums and Communities: • Stack Overflow: A community where you can ask questions related to networking and data communication. • Reddit: Subreddits like r/networking and r/NetworkingForBeginners can be helpful for discussions and learning from others' experiences.
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Assessment Plan:

Component of Evaluation	Description	Code	Weightage %
Continuous Internal Evaluation	Mid Term	CT	15%
	Seminar/Viva-Voce/Quiz/Home Assignment	S/V/Q/HA	10%
Attendance	A minimum of 75% Attendance is required to be maintained by a student to be qualified for taking the End Semester examination. The dispensation of 25% includes all types of leaves. including medical leaves.	A	5%
End Semester Examination	End Semester Examination	ESE	70%
Total			100%

Abbreviations: CT: Class Test, HA: Home Assignment, S/V/Q: Seminar/Viva/Quiz, ESE: End Semester Examination; A: Attendance

Course Articulation Matrix (Mapping of COs with POs)

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	3	3	2	3			2	2	3		1	2	3	2
CO2	3	3	3	3	1				2	2			3	2	1
CO3	3	3	3	2	3	1		2	3	2	2	1	3	2	2
CO4	3	3	3	3	2			1	2	2		2	3	3	3
CO5	3	3	3	3	2	1		1	2	2	1	2	2	2	2

1: strongly related, 2: moderately related and 3: weakly related

COMMUNICATION SKILLS – VI**Course Code: BCU 641****Credit Units: 01****Total Hours: 15****Course Objective:**

- The main emphasis of this course is to enable students to learn the dynamics of social communication and to demonstrate the ability to learn the nuances of informal communication.

Prerequisites: NIL

Prerequisites: Nil

Course Contents / Syllabus:				
1.	Module I: Social Communication Essentials			30% Weightage
	• Small talk/Building rapport • Expand social and Corporate Associations • Informal Communication: Grapevine, Chat			
2.	Module II: Workplace Interpersonal Skills			25% Weightage
	• Understanding Social Communication in Workplace environment. • Employee feedback: Assess employee performance and satisfaction. • Simulation ➤ Humour in Communication-Use of ‘Puns’ ➤ Entertainment and Communication (Infotainment) • Infotainment and Social Media • Entertainment in Journalism ➤ Social Networking			
3.	Module III: Verbal Ability			35% Weightage
	• Comprehension • Antonyms, Synonyms • Idioms & Phrases & Analogy • Sentence Order • Active and Passive Voice • Error Sorting			
4.	Module IV: Prose			10% Weightage
	• Secret of Socrates - Dale Carnegie • My Financial Career-Stephen Leacock • The Luncheon - W. Somerset Maugham • The National Flag - Jawahar Lal Nehru All the four stories will be discussed in one class One Long Question will be set in the Exam from the Text			
5.	Student Learning Outcomes:			
	• To communicate contextually in specific personal and professional situations with courtesy. • To inject humour in their regular interactions. • To strengthen their creative learning process through individual expression and collaborative peer activities.			
6.	Pedagogy for Course Delivery:			
	• Workshop • Group Discussions • Presentations • Lectures			
7.	Assessment/ Examination Scheme:			
	Theory L/T (%)	Lab/Practical/Studio (%)	End Term Examination	
	100%	NA	70%	
	Theory Assessment (L&T):			
	Components (Drop down)	CIE	Mid Sem	Attendance
Weightage (%)	10%	15%	5%	70%

Text & References:

- Krizan, Merrier, Logan & Williams. *Effective Business Communication*, New Delhi: Cengage, 2011
- Communication and Organizational Culture*. Keyton. Joann. Sage Publications
- Social Communication (Frontiers of Social Psychology)*. Fiedler, Klaus. Psychology Press
- Cyberpunks: Freedom and the Future of the Internet*. Assange, Julian Assange. OR Books.

Course Code: BSU 643**Credit Units: 01****Total Hours: 15****Course Objective:**

- To develop an understanding the concept of stress its causes, symptoms and consequences.
- To develop an understanding the consequences of the stress on one's wellness, health, and work performance.

Course Contents:**Module I: Stress****(03 Hours)**

- Meaning & Nature
- Characteristics
- Types of stress

Module II: Stages and Models of Stress**(03 Hours)**

- Stages of stress
- The physiology of stress
- Stimulus-oriented approach.
- Response-oriented approach.
- The transactional and interactional model.
- Pressure – environment fit model of stress.

Module III: Causes and symptoms of stress**(03 Hours)**

- Personal
- Organizational
- Environmental

Module IV: Consequences of stress**(03 Hours)**

- Effect on behavior and personality
- Effect of stress on performance
- Individual and Organizational consequences with special focus on health

Module V: Strategies for stress management**(03 Hours)**

- Importance of stress management
- Healthy and Unhealthy strategies
- Peer group and social support
- Happiness and well-being

Course Outcomes:

- Student will able demonstrate thorough understanding of stress and its effects
- Student will able to learn various coping strategies to deal stress effectively so to overcome the consequences and impact of stress on their health and wellbeing, ultimately it will enhance their performance.

Examination Scheme:

Evaluation Components	Attendance	Journal of Success (JOS)	Social Awareness Program (SAP) SAP Report/SAP Presentation	End Semester Exam	Total
Weightage (%)	5	10	15	70	100

Text & References:

- Blonna, Richard; Coping with Stress in a Changing World: Second edition
- Pestonjee, D.M, Pareek, Udai, Agarwal Rita; Studies in Stress And its Management
- Pestonjee, D.M.; Stress and Coping: The Indian Experience

PROBLEM SOLVING THROUGH AI&ML

Course Code: CSA 601**Credit Units: 03****Total Hours: 45****Course Objective:**

The main objective is to teach utility of basic methods of Artificial Intelligence and Machine learning in various case studies in industries, including:, planning, search, localization of errors, tracking, prediction and control of the system.

Course Contents:**Module I: Introduction: (9 Hours)**

Machine Learning Applications across Industries - Healthcare, Retail, Financial Services, Manufacturing, Hospitality

Module II: Churn Analysis and Fraud Detection: (9 Hours)

Churn Analysis and Prediction (Survival Modelling): Churn Prediction, Credit card Fraud Analysis: Imbalanced Data, Neural Network

Module III: Handling Text Data: (9 Hours)

Handling Text Data, Bag-of-words, Regular Expressions, Sentence Splitting and Tokenization, Punctuations and Stop words, Incorrect spellings, Properties of words and Word cloud, Sentiment Analysis (Case Study)

Module IV: Forecasting and Prediction: (9 Hours)

Forecasting models, Trend analysis, Cyclical and Seasonal analysis, Smoothing; Moving averages; Box-Jenkins, Holt-winters, Auto-correlation; ARIMA, Examples: Applications of Time Series in financial markets

Module V: Recommender Systems: (9 Hours)

Recommendation Systems and Collaborative filtering, User based, Item Based, Singular value decomposition-based recommenders, Customer Segmentation and Value, Social Networking Analysis

Course Outcomes:

At the end of this course, students will learn:

- To study concepts of various real-time case studies with respect to machine learning.
- To understand the utility of ML techniques in industrial use cases.

Examination Scheme:

Components	A	CT	S/V/Q/HA	EE
Weightage (%)	5	15	10	70

CT: Class Test, HA: Home Assignment, S/V/Q: Seminar/Viva/Quiz, EE: End Semester Examination; A: Attendance

Text & References:**Text:**

- Machine Learning by Saikat Dutt , Subramanian Chandramouli, Pearson Education; First edition (1 October 2018)
- Chandra S.S.V, Artificial Intelligence and Machine Learning, Prentice Hall India Learning Private Limited; 1 edition (2014)

References:

- Tom M. Mitchell, —Machine Learning, McGraw-Hill Education (India) Private Limited, 2013.
- Ethem Alpaydin, —Introduction to Machine Learning (Adaptive Computation and Machine Learning), The MIT Press 2004.
- Stephen Marsland, —Machine Learning: An Algorithmic Perspective, CRC Press, 2009.
 - Christopher M. Bishop, Pattern Recognition and Machine Learning.

DATA VISUALIZATION

Course Code: CSD 601

Credit Units: 03
Total Hours: 30

Course Objective:

The course will introduce students to the fundamental of data Visualizations. An understanding of the key techniques and theory used in visualization, including data models, graphical perception and techniques for visual encoding and interaction.

Course Contents:

Module I: Defining Plots: (8 Hours)

bar, line, and stacked charts, axis lengths and limits , plot line styles, properties, and format , Colour Background, colour Setting ticks, labels, and grids histograms , bar plot ,pie chart, scattered plot, using matplotlib and ggplot2.

Module II: Advance Plots: (6 Hours)

Subplots, Data Table and Customizations, transparency and size of axis labels, adding aesthetics, grids, contour plots, Polar plots, colour theory, using matplotlib and ggplot2.

Module III: Plotting With Maps and Images: (8 Hours)

Probability Processing images with PIL, plotting images, Base map, leaflet.js, Captcha Images using matplotlib and ggplot2.

Module IV: Choosing Right Plots: (8 Hours)

Unconstrained optimization; spectrograms, stem plot, streamlines of vector flow, colormaps, scatter plots and histograms, cross-correlation between two variables, Importance of autocorrelation, Latex, Power BI, Tableau.

Course Outcomes:

Upon successful completion of this unit, students should be able to:

- Plot data with right metrics;
- Selection of optimal plots;
- Demonstrate enhanced analytics report,

Examination Scheme:

Components	A	CT	S/V/Q/HA	ESE
Weightage (%)	5	15	10	70

A: Attendance, CT: Class Test, S/V/Q/HA: Seminar/Viva/Quiz/ Home Assignment, EE: End Semester Examination;

Text book & References:

Text books:

- Cole Nussbaumer Knaflitz (2015). Storytelling with Data: A Data Visualization Guide for Business Professionals
- Scott Berinato (2016). Goodcharts

Reference Books:

- Robert Collins (2018). Introduction to data visualization with python R, Table

DATA VISUALIZATION LAB

Course Code: CSD 621

Credit Unit: 01
Total Hours : 20

Course Objective:

The course will introduce students to the fundamental of data Visualizations. An understanding of the key techniques and theory used in visualization, including data models, graphical perception and techniques for visual encoding and interaction.

Software Requirements: Python 3.6, Anaconda IDE with Spider

Consider this given Data-set for which we will be plotting different charts for experiments 1- 5:

S. No.	Empid	Gender	Age	Sales	BMI	Income
1	E1	M	32	123	NORMAL	350
2	E2	M	35	114	OVERWEIGHT	450
3	E3	F	34	135	OBESITY	170
4	E4	F	42	139	UNDERWEIGHT	190
5	E5	F	26	117	NORMAL	90
6	E6	M	38	121	OBESITY	140
7	E7	F	44	133	OVERWEIGHT	125
8	E8	M	24	140	UNDERWEIGHT	95
9	E9	M	36	133	OVERWEIGHT	200
10	E10	F	40	133	OBESITY	310

List of experiments/demonstrations:

1. Write python code to implement a histogram from above code.
2. Write python code to implement a Column chart from above code
3. Write python code to implement a Box Plot Chart from above code
4. Write python code to implement a Pie Chart from above code
5. Write python code to implement a Scatter Plot from above code
6. Using pandas module write a data frame to csv file
7. Write python code to implement Mandelbrot Fractal Set visualization in Python
8. Write python code to plot Fibonacci spiral fractal using Turtle
9. Implement KDE (Kernel Density Estimate) Plot Visualization with Pandas and Seaborn
10. Implement Violin Plot for Data Analysis.

Course Outcomes:

Upon successful completion of this unit, students should be able to:

- Plot data with right metrics;
- Selection of optimal plots;
- Demonstrate enhanced analytics report,

Examination Scheme:

IA			EE			
A	PR	Practical Based Test	Major Experiment	Minor Experiment	LR	Viva
5	10	15	35	15	10	10

Note: IA –Internal Assessment, EE- External Exam, A- Attendance PR- Performance, LR – Lab Record, V – Viva.

Text Books:

- Cole Nussbaumer Knaflic (2015). Storytelling with Data: A Data Visualization Guide for Business Professionals
- Scott Berinato (2016). Goodcharts

Reference Books:

- Robert Collins (2018). Introduction to data visualization with python R, Table



Syllabus

Programme Name	B. Tech. (CSE)		BASIC	Batch	2025-2029
Course Code	CSC 601	Course Name	Web Application Security And Penetration Testing Design	Semester	VI

Credits (Total)	L	T	P	Marks (Internal/External)		Contact Hours (per week)	Independent Study Hour (per week)
3	3	0	0	30	70	3	3

Course Objective:

The course will cover the concepts involved in web application development. The course will introduce to various vulnerabilities in web applications and their mitigation techniques.

Course Alignment with UNSDG:

The Course aims to fulfill the United Nations Sustainable Development Goals SDG 4 (Quality Education)

Course outcomes:

After completion of course, the student will be able to:

CO-1	Apply client-side web development to design interactive front-end web user interfaces.
CO-2	Understand server-side web application concepts to develop back-end web server application
CO-3	Identify and mitigate various client-side web application security vulnerabilities
CO-4	Identify and mitigate various server-side web application security vulnerabilities

Teaching Pedagogy:

T1	Activity based learning through lab experiments like Dissection, specimen observation Power Point Presentations and white board teaching
T2	Class Test, Assignments, Flip Class/ Seminars, Quiz, Oral Viva-voce examination

Assessment Tools

AT1-1	Quiz	AT1-6	Poster
AT1-2	Midterm Exams	AT1-7	Oral Viva-voce examination
AT1-3	Flip Class	AT1-8	
AT1-4	Seminar Presentation	AT1-9	
AT1-5	Assignments		

Prerequisites: Basics of Cyber security and Computer Networking

Module	Descriptors/Topics	Hours	Assessment tools
I	Introduction: Web application development – Introduction - Architecture – Client-side technologies and frameworks – HTML – CSS – Java Script - Ajax/Fetch - Data interchange formats – XML, JSON. Server-side scripting and technologies - development – technologies - Handling client requests – Database connectivity – Sessions – Cookies	9	Quiz Mid-term Exam Assignment
II	Security threats and Vulnerabilities: Web application vulnerabilities – Client-side Vulnerabilities - Cross Site Scripting (XSS) - Cross Site Request Forgery (CSRF) - Cross-origin resource sharing (CORS) – Clickjacking; Server-side Vulnerabilities - SQL injection - OS command injection - Directory traversal - Authentication - Server-side request forgery (SSRF).	9	Mid-Term Quiz Assignment
III	Protocol engineering: Experiments with open-source network simulators (NS2 and NS3): Installation and configuration, Creation of network topology and understanding of packet switched network, Simulation and visualization of different types of traffic—congestion controlled, and non-congestion controlled, Trace analysis and visualization of protocol dynamics.	9	Mid-Term Oral Viva- voce examination Seminar Presentation
IV	Implementing network security Protocols: Simulation with active queue management schemes, Simulation and visualization of attacks (e.g. IP spoofing and reflection attacks). Socket programming: implementation of IP spoofing and reflection DDoS attacks. Linux Kernel: Familiarization with Linux kernel protocol implementation (TCP/IP) implementation, Tracing and debugging of Linux Kernel TCP/IP source code, Kernel modification and recompilation, Implementation of a non- attack in Linux kernel Network Emulation and testbeds: Network emulation and traffic control using tc (traffic control), dummynet and other advanced tools, Familiarization with advanced testbed technologies (e.g. Emulab, DETER and PlanetLab, etc.), formal verification of protocol: SPIN, UPPAL.	9	Quiz Assignment Industrial Visit Report Seminar Presentation
V	Design and Implementation of Penetration Testing: Causes of Vulnerabilities, Penetration Testing Tools, Use external penetration testers to find problems, Planning and reconnaissance, scanning of ports, Penetration testing methods, Feedback results to the defect management and mitigation system, Scheduling of periodic penetration tests for application coverage, use external penetration testers to perform deep-dive analysis, Customize penetration testing tools.	9	Quiz Assignment Industrial Visit Report Seminar Presentation
	Total Teaching Hours	45	

Additional Resources

A. Value addition to course content/ Skill enhancement content

- a. <https://www.cybrary.it/course/cyber-security-for-beginners/>

B. Remedial classes for slow learners

As per the AUMP SOP for Slow & Advanced Learn

Suggested reading:

1. Textbooks: W. Stallings, Network Security Essentials: Applications and Standards, 6th Edition, Pearson Prentice Hall, 2016. - Bryan Sullivan and Vincent Liu, Web Application Security, A Beginner's Guide, McGraw-Hill Education, 2012. - Kaufman, R. Perlman and M. Speciner, Network Security: Private Communication in a Public World, 2nd Edition, Prentice Hall PTR, 2002
2. Reference books Internet of Things: A Hands-On Approach by Arshdeep Bahga, Vijay Madisetti
3. Suggested e- resources (Websites/e-books) https://attack.mitre.org/

Assessment Plan:

Component of Evaluation	Description	Code	Weightage %
Continuous Internal Evaluation	Mid Term	CT	15%
	Home Assignment/Seminar/ Viva-Voce/Quiz	S/ HA/ V/Q	10%
Attendance	A minimum of 75% Attendance is required to be maintained by a student to be qualified for taking the End Semester examination. The dispensation of 25% includes all types of leaves. including medical leaves.	A	5%
End Semester Examination	End Semester Examination	ESE	70%
Total			100%

Abbreviations: CT: Class Test, HA: Home Assignment, S/V/Q: Seminar/Viva/Quiz, ESE: End Semester Examination; A: Attendance

Course Articulation Matrix (Mapping of COs with POs)

Course Outcomes	Correlation with POs												Correlation with PSOs		
	P O 1	P O 2	P O 3	P O 4	P O 5	P O 6	P O 7	P O 8	P O 9	P O 10	P O 11	P O 12	P S O 1	P S O 2	P S O 3
CO1	1	1	1	2	3						2	3	2	2	
CO2	1	1	1											2	
CO3	1	1	1	3							2		2	1	
CO4	1	2	1	3									2	1	

1: weakly related, 2: moderately related and 3: strongly related



Syllabus

Programme Name	B. Tech(CSE)		APPLIED	Batch	2025-2029
Course Code	CSC 621	Course Name	Web Application Security And Penetration Testing Design Lab	Semester	VI

Credits (Total)	L	T	P	Marks (Internal/External)		Contact Hours (per week)	Independent Study Hour (per week)
1	0	0	2	30	70	2	2

Course Objective:

This course aims to delve into the fascinating world of securing web applications against malicious attacks and conducting penetration tests to assess their security posture.

Course Alignment with UNSDG:

The Course aims to fulfill the United Nations Sustainable Development Goals SDG 4 (Quality Education) and SDG 9: Industry, Innovation and Infrastructure

Course outcomes:

After completion of course, the student will be able to:

CO-1	Familiarize the open-source network simulators and its experiments.
CO-2	Understand the protocol dynamics, simulation with active queue management schemes and visualization of attacks.
CO-3	Familiarize with Linux Kernel Protocol implementation, kernel modification and recompilation.
CO-4	Explore formal verification of protocols, network emulation, and testbed technologies.

Teaching Pedagogy:

T1	Activity based learning through lab experiments like Dissection, specimen observation Power Point Presentations and white board teaching
T2	Class/Seminars Quiz/ Assignments/Case studies

Lab Plan

Lab Session	Practical/Lab Exercise	Mapped CO	Mode of Assessing CO
Lab 1	Install and configure a virtual lab environment using tools like Virtual Box or VMware. Set up a target web application server (e.g., DVWA - Damn Vulnerable Web Application) and a separate machine for running security testing tools	CO1	Viva/Quiz/Practical Performance
Lab 2	Use reconnaissance techniques such as DNS enumeration, WHOIS lookup, and spidering to gather information about the target web application, including its architecture, technologies used, and potential entry points.	CO1	Viva/Quiz/Practical Performance
Lab 3	Utilize automated vulnerability scanning tools like Nessus, OpenVAS, or Nikto to scan the target web application for known vulnerabilities, misconfigurations, and common security issues.	CO1	Viva/Quiz/Practical Performance
Lab 4	Exploit discovered vulnerabilities in the web application using tools such as Metasploit, SQLmap, or Burp Suite. Practice exploiting common vulnerabilities like SQL injection, cross-site scripting (XSS), and command injection to gain unauthorized access or escalate privileges.	CO2	Viva/Quiz/Practical Performance
Lab 5	Attempt to bypass authentication mechanisms implemented in the web application using techniques like brute force attacks, credential stuffing, or session manipulation.	CO2	Viva/Quiz/Practical Performance
Lab 6	Test the effectiveness of access control mechanisms by attempting to access restricted	CO2	Viva/Quiz/Practical Performance

	areas or perform unauthorized actions within the web application. Explore different user roles and permissions to identify potential access control vulnerabilities.		
Lab 7	Test the input validation mechanisms of the web application to identify vulnerabilities such as SQL injection, XSS, and command injection. Submit malicious input and analyze how the application handles and sanitizes user input.	CO3	Viva/Quiz/Practical Performance
Lab 8	Assess the security of session management mechanisms ,including session cookies, tokens, and session fixation vulnerabilities. Attempt to hijack or manipulate user sessions to gain unauthorized access to the application.	CO3	Viva/Quiz/Practical Performance
Lab 9	Explore client-side attacks such as cross-site scripting (XSS),cross-site request forgery (CSRF), and clickjacking. Craft malicious payloads and demonstrate their impact on user browsers.	CO3	Viva/Quiz/Practical Performance
Lab 10	Test the security of APIs used by the web application to communicate with backend services or third-party applications.Identify potential vulnerabilities such as insecure API endpoints, parameter tampering, or lack of authentication / authorization controls.	CO3	Viva/Quiz/Practical Performance
Lab 11	Document all findings, including identified vulnerabilities, their severity, and potential impact on the web application's security. Provide detailed	CO4	Viva/Quiz/Practical Performance

	recommendations for remediation and prioritize		
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Suggested reading:

1. Textbooks: W. Stallings, Network Security Essentials: Applications and Standards, 6th Edition, Pearson Prentice Hall, 2016. - Bryan Sullivan and Vincent Liu, Web Application Security, A Beginner's Guide, McGraw-Hill Education, 2012. - Kaufman, R. Perlman and M. Speciner, Network Security: Private Communication in a Public World, 2nd Edition, Prentice Hall PTR, 2002
2. Reference books Internet of Things: A Hands-On Approach by Arshdeep Bahga, Vijay Madisetti
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Assessment Plan:

Component of Evaluation	Description	Code	Weightage %
Continuous Internal Evaluation	Mid Term	CT	15%
	Viva-Voce/Assignments/Practical records	V/P/PR	10%
Attendance	A minimum of 75% Attendance is required to be maintained by a student to be qualified for taking the End Semester examination. The dispensation of 25% includes all types of leaves. including medical leaves.	A	5%
End Semester Practical Examination	End Semester Practical Examination	ESE	70%
Total			100%

Abbreviations: CT: Class Test, V/A/PR: Viva/Assignments/Practical records, ESE: End Semester Practical Examination; A: Attendance

Course Articulation Matrix (Mapping of COs with POs)

Course Outcomes	Correlation with POs												Correlation with PSOs		
	P O 1	P O 2	P O 3	P O 4	P O 5	P O 6	P O 7	P O 8	P O 9	P O 10	P O 11	P O 12	P S O 1	P S O 2	P S O 3
CO1	1									3			3		
CO2	3	2		3				2					2		2
CO3	3	2		2						3			2		
CO4	3							2					2		2

1: weakly related, 2: moderately related and 3: strongly related

IOT APPLICATIONS DESIGN

Course Code: CSI 601

Credit Units: 03
Total Hours: 30

Course Objectives:

- Ability to understand the coding methodology for IOT.
- Understand the software framework for IoT
- Ability to understand the hardware to hardware communication and hardware to software communication
- Understand the various types of advance input/output interface.

Course Contents:

Module I: Introduction: (8 Hours)

Introduction – History - Creative Coding Platforms - Open Source Platforms – PIC - Arduino, Sketch, Raspberry Pi, Iterative coding methodology – Python Programming - Mobile phones and similar devices - Arm Devices - Getting used to Arduino - Sensor Characterization: Safety, Basic Electronics (circuit theory, measurements, parts identification) Sensors and Software: Understanding Processing Code Structure, variables and flow control, Interfacing to the Real World

Module II: Software Frameworks: (6 Hours)

Software: open Frameworks as our IDEs - “Arduino” Languages - Hardware: Desktop / Laptop / Raspberry Pi - How to approach a programming problem? Representing “reality” with computers. Digital vs. Analog circuits, audio, communication, etc. Analog to Digital Conversion (ADC) - Digital to Analog Conversion (DAC) – Microcontrollers, Python Programming for IoT.

Module III: Hardware Communication: (8 Hours)

Communication – Serial& Parallel - Hardware to Hardware Communication - I2C/IIC (Inter-Integrated Circuit) - SPI (Serial Peripheral Interface) – Serial UART Communication - Introduction to the command line – git/github. Introduction to Programming: A comparative studio between Arduino and open Frameworks - Arduino-compatible Microcontrollers Sensors and Actuators

Module IV: Advanced I/O Interfacing: (8 Hours)

Advanced I/O – open Frameworks: Live Network feeds (push and pull) - Data persistence (saving data and preferences) - Database interface (MySQL, sqLite, XML, PHP/Web) - Arduino: Wired/Wireless Networking

UNIT V - IoT, FUTURE AND PERSPECTIVES (6 hours) Talking to the cloud: Baby steps to Internet of Things, TCP/IP and UDP - Building peer to peer communication system using Bluetooth & WiFi.

Course Outcomes:

The student will learn

- Understand the concept of IOT Open Source Platform.
- Understand software platform for IOT.
- How use & Access hardware communication for IOT.
- How can send and stored data in database which is sense by sensor from remote location.

Examination Scheme:

Components	A	CT	S/V/Q/HA	EE
Weightage (%)	5	15	10	70

A: Attendance, CT: Class Test, S/V/Q/HA: Seminar/Viva/Quiz/ Home Assignment, EE: End Semester Examination;

Text & References:

- Internet of things programming projects by colin dow: Build IoT solution with Raspberry Pi 3 and Python
- Hands on Internet of Things with Blynk by Pradeeka Seneviratne
- Industrial Internet Application Development by Alena Traukina, Jayant Thomas

IOT APPLICATIONS DESIGN LAB

Course Code: CSI 621

Credit Unit: 01

Total Hours: 20

Course Objectives:

- Ability to understand the coding methodology for IOT.
- Understand the software framework for IoT
- Ability to understand the hardware to hardware communication and hardware to software communication
- Understand the various types of advance input/output interface.

Programs List:

1. Write a program for Arduino board which takes input from infra red sensor and display count on 16 X 2 LED screen: **(2 Hours)**
2. Write a program to send SMS on a mobile phone from Arduino board using GSM module: **(2 Hours)**
3. Write a program to receive call from any mobile using Android board and GSM module: **(2 Hours)**
4. Write a program to sense temperature at remote location using Arduino board and temperature sensor: **(2 Hours)**
5. Write a program for Arduino board to sense moisture at some remote location using moisture sensor: **(2 Hours)**
6. Write a program for Arduino board to create smart burglar alarm using suitable sensors: **(2 Hours)**
7. Design a circuit board for water level detector system using suitable microcontroller and sensors: **(2 Hours)**
8. Design and program a complete smart home system using appropriate sensors, actuators and microcontroller unit for automatic regulation of home electrical appliances like room bulbs, ceiling fans etc: **(2 Hours)**
9. Consider a customer who wants automatic water pump system for his home such that the system must be able to switch on and off water pump automatically without his intervention. Design and program the complete system using appropriate sensors, actuators and microcontroller unit: **(2 Hours)**
10. Consider customer want a dedicated small power web server for his home automation system such that he/she can communicate between his/her various home appliances. Design and program a complete smart home dedicated web server system using appropriate sensors, actuators and microcontroller unit: **(2 Hours)**

Course Outcomes:

The student will learn

- Understand the concept of IOT Open Source Platform.
- Understand software platform for IOT.
- How use & Access hardware communication for IOT.
- How can send and stored data in database which is sense by sensor from remote location.

Examination Scheme:

IA			EE			
A	PR	Practical Based Test	Major Experiment	Minor Experiment	LR	Viva
5	10	15	35	15	10	10

Note: IA –Internal Assessment, EE- External Exam, A- Attendance PR- Performance, LR – Lab Record, V –Viva.

Text & References:

- Internet of things programming projects by colin dow: Build IoT solution with Raspberry Pi 3 and Python.
- Hands on Internet of Things with Blynk by Pradeeka Seneviratne.
- Industrial Internet Application Development by Alena Traukina, Jayant Thomas.

MINOR PROJECT**Course Code: NMP 660****Credit Units: 02****Course objectives:**

The objective of Minor project is to enable the student to take up investigative study in the broad field of Engineering, either fully theoretical/practical or involving both theoretical and practical work to be assigned by the Department on an individual basis or two/three students in a group, under the guidance of a Supervisor.

Methodology:

Topics of project are to be based on the latest trends, verifying engineering concepts /principals and should involve elementary research work. The projects may involve design, fabrications, testing, computer modeling, and analysis of any engineering problem. On completion of the practical training the students are to present a report covering various aspects learnt by them and give a presentation on same.

Course Outcomes:

At the end of the course, students will demonstrate the ability to:

- 1 Conceive a problem statement either from rigorous literature survey or from the requirements raised from need analysis.
- 2 Design, implement and test the prototype/algorithm in order to solve the conceived problem.
- 3 Write comprehensive report on mini project work.

Examination Scheme:

Literature study/ Fabrication/ Experimentation	40
Written Report	20
Viva	15
Presentation	25
Total	100